Tutorial-10

$\boxed{w_i = 1/N}$

$N \Rightarrow$ no. of training samples

$w_i \Rightarrow$ weight of sample i

Sample	Age	Income	Credit Score	Loan Approved
S1	25	40	650	Yes
S2	30	60	700	Yes
S3	45	35	600	No
S4	50	30	580	No
S5	28	35	720	Yes

$$h(x) = \begin{cases} +1 & x < \text{threshold} \\ -1 & x \geq \text{threshold} \end{cases}$$

Normalize weights

$$w_i = \frac{w_i^{new}}{\sum w_i^{new}}$$

weight error

$$e = \sum_{i=1}^{n} w_i \, I(y_i \neq h(x_i))$$

$I \Rightarrow$ indicator function

$$I \Rightarrow \begin{cases} 0, & y = h(x) \\ 1, & y \neq h(x) \end{cases}$$

weight updation

$$w_i^{new} = w_i^{old} \cdot e^{-\alpha y_i h(x_i)}$$

$\Rightarrow w_1 = 1/5 = 0.2$

$\Rightarrow w_2 = 1/5 = 0.2$

$\Rightarrow w_3 = 1/5 = 0.2$

$\Rightarrow w_4 = 1/5 = 0.2$

$\Rightarrow w_5 = 1/5 = 0.2$

(threshold = 45)

⇒

Age	h(x)	y	I
25	+1	+1	0
30	+1	+1	0
45	-1	-1	0
50	-1	-1	0
28	+1	+1	0

~~⇒ e =~~ error

$e = 0.2(0) + 0.2(0) + 0.2(0) + 0.2(0) + 0.2(0)$

$\Rightarrow e = 0$

$$\alpha = \frac{1}{2}\ln\left(\frac{1-e}{e}\right) = \frac{1}{2}\ln\left(\frac{1-0}{0}\right) = \frac{1}{2}\ln(\infty)$$

$\alpha = \infty$

weight update

$$w_1^{new} = (0.2)\left(e^{-(\infty)(+1)(+1)}\right) = 0$$

$\Rightarrow w_2^{new} = 0,\ w_3^{new} = 0,\ w_4^{new} = 0,\ w_5^{new} = 0$

Normalize weights

$$w_1 = \frac{w_1^{new}}{\sum w_i^{new}} = \frac{0}{0}$$

(threshold = 28)

⇒

Age	h(x)	y	I
25	+1	+1	0
30	-1	+1	1
45	-1	-1	0
50	-1	-1	0
28	-1	+1	1

error

$e = 0.2(0) + 0.2(1) + 0.2(0) + 0.2(0) + 0.2(1) = 0.4$

$$\alpha = \frac{1}{2}\ln\left(\frac{1-0.4}{0.4}\right) = \frac{1}{2}\ln(3/4)$$

$$\alpha = \frac{1}{2}\ln\left(3/2\right) \approx 0.202$$

<u>weight update</u>

$$w_1^{new} = (0.2)\left(e^{-0.202(+1)(+1)}\right)$$

$$= 0.16$$

$$w_2^{new} = (0.2)\left(e^{-0.2(-1)(+1)}\right)$$

$$= 0.24$$

$$w_3^{new} = (0.2)\left(e^{-0.2(-1)(-1)}\right)$$

$$= 0.16$$

$$w_4^{new} = 0.16$$

$$w_5^{new} = 0.24$$

<u>Normalize weights</u>

$$w_p = \frac{w_p^{new}}{\sum w_p^{new}}$$

$$w_1 = \frac{0.16}{0.96} = 0.166$$

$$w_3 = w_4 = 0.166$$

$$w_2 = \frac{0.24}{0.96} = 0.25$$

$$w_5 = 0.25$$

⇒

Income	$h(x)$	y	I
40	-1	+1	1
60	-1	+1	1
35	-1	-1	0
30	+1	-1	1
55	-1	+1	1

(threshold = 35)

error

$$e = 0.166(1) + 0.25(1) + 0.166(0) + 0.166(1) + 0.25(1)$$

$$= 0.832$$

$$\alpha = \frac{1}{2}\ln\left(\frac{1-e}{e}\right) = \frac{1}{2}\ln\left(\frac{0.168}{0.832}\right) = \frac{1}{2}(-1.59)$$

$$\alpha = -0.79$$

weight update

$$w_1^{new} = (0.166)\left(e^{-(-0.79)(-1)(+1)}\right)$$

$$= 0.166 \times 0.45 = 0.075$$

$$w_2^{new} = 0.25\left(e^{-(-0.79)(-1)(+1)}\right)$$

$$= 0.113$$

$$w_3^{new} = (0.166)\left(e^{-(-0.79)(-1)(-1)}\right)$$

$$= 0.365$$

$$w_4^{new} = 0.075$$

$$w_5^{new} = 0.113$$

$$\sum w_i^{new} = 0.741$$

Normalize weights

$$w_1 = \frac{0.075}{0.741} = 0.101$$

$$w_2 = \frac{0.113}{0.741} = 0.152$$

$$w_3 = \frac{0.365}{0.741} = 0.492$$

$$w_4 = \frac{0.075}{0.741} = 0.101$$

$$w_5 = \frac{0.113}{0.741} = 0.152$$

→

Credit Score	$h(x)$	y	I
650	-1	+1	1
700	-1	+1	1
600	-1	-1	0
580	+1	-1	1
720	-1	+1	1

(threshold = 600)

error

$$e = 0.101(1) + 0.152(1) + 0.492(0) + 0.101(1) + 0.152(1)$$

$$= 0.506$$

$$\alpha = \frac{1}{2}\ln\left(\frac{1-e}{e}\right) = \frac{1}{2}\ln\left(\frac{0.494}{0.506}\right) = -0.012$$

weight update

$$w_1^{new} = 0.101\left(e^{-(-0.012)(-1)(+1)}\right)$$

$$= 0.099$$

$$w_2^{new} = 0.152\left(e^{-(-0.012)(-1)(+1)}\right)$$

$$= 0.15$$

$$w_3^{new} = 0.492\left(e^{-(-0.012)(-1)(-1)}\right)$$

$$= 0.497$$

$$w_4^{new} = 0.099$$

$$w_5^{new} = 0.15$$

$$\Sigma w_p = 0.995$$

Normalize weights

$$w_1 = \frac{0.099}{0.995} = 0.0994$$

$$w_2 = \frac{0.15}{0.995} = 0.1507$$

$$w_3 = \frac{0.497}{0.995} = 0.4994$$

$$w_4 = 0.0994$$

$$w_5 = 0.1507$$